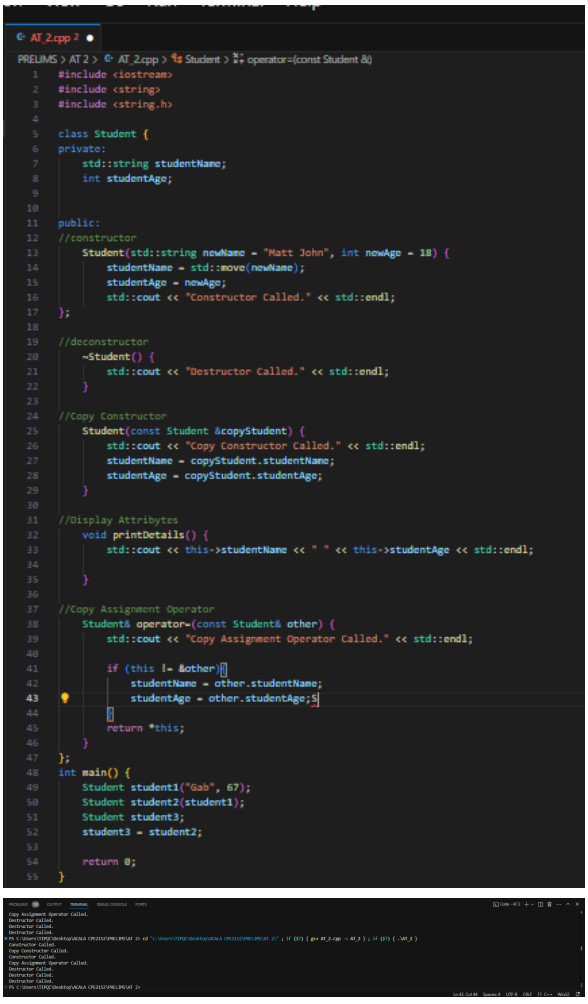
COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	07/17/26	**DATE SUBMITTED:**	07/17/26
NAME:	MATT JOHN ACALA	**INSTRUCTOR:**	Engr. Jimlord

ACTIVITY NO. 2 **Arrays, Pointers and Dynamic Memory Allocation**

1. PROCEDURE OUTPUT

Screenshot	
Observation	Base on my observation some of the code have some errors for example the copy assignment operator it has a delete function but does not have any heap so its dangerous.

```
#include <iostream>
#include <string>
#include <string.h>

class Student {
private:
    std::string studentName;
    int studentAge;


public:
//constructor
    Student(std::string newName = "Matt John", int newAge = 18) {
        studentName = std::move(newName);
        studentAge = newAge;
        std::cout << "Constructor Called." << std::endl;
};

//deconstructor
    ~Student() {
        std::cout << "Destructor Called." << std::endl;
    }

//Copy Constructor
    Student(const Student &copyStudent) {
        std::cout << "Copy Constructor Called." << std::endl;
        studentName = copyStudent.studentName;
        studentAge = copyStudent.studentAge;
    }

//Display Attribytes
    void printDetails() {
        std::cout << this->studentName << " " << this->studentAge << std::endl;

    }

//Copy Assignment Operator
    Student& operator=(const Student& other) {
        std::cout << "Copy Assignment Operator Called." << std::endl;

        if (this != &other){
            studentName = other.studentName;
            studentAge = other.studentAge;}
        return *this;
    }
};
int main() {
    Student student1("Gab", 67);
    Student student2(student1);
    Student student3;
    student3 = student2;

    return 0;
}
```

Table 2.1 Initial Drive Program

Screenshot	
Observation	It still runs successfully since i fixed the error earlier

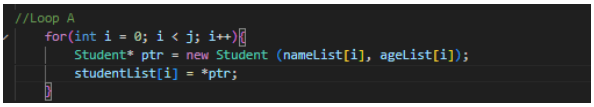
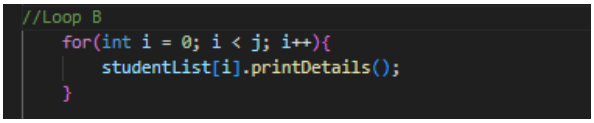
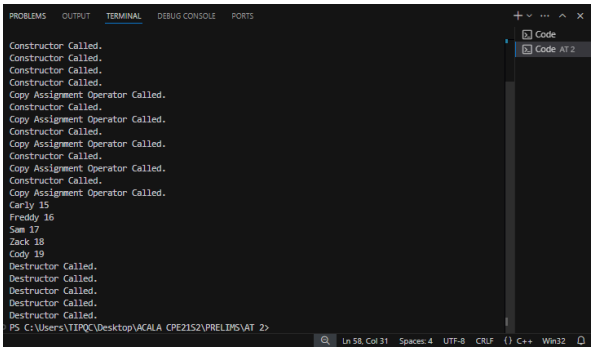
```
int main() {
    const size_t j = 5;

    Student studentList[j] = {};
    std::string nameList[j] = {"Carly", "Freddy", "Sam", "Zack", "Cody"};
    int ageList[j] = {15, 16, 17, 18, 19};

    return 0;
}
```

```
PS C:\Users\TIPQC\Desktop\ACALA CPE21S2> cd "c:\Users\TIPQC\Desktop\ACALA CPE21S2\PRELIMS\AT 2\" ; if ($?) { g++ AT_2.cpp -o AT_2 } ; if ($?) { .\AT_2 }
Constructor Called.
Copy Constructor Called.
Constructor Called.
Copy Assignment Operator Called.
Destructor Called.
Destructor Called.
Destructor Called.
PS C:\Users\TIPQC\Desktop\ACALA CPE21S2\PRELIMS\AT 2>
```

Table 2.2 Modified Driver Program with Student Lists

Loop A	
Observation	So for these it just use the constructor and copy assignment constructor for the 5 students
Loop B	
Observation	While for this it use the print attributes of the class to print the student name and age and also used the constructor. Lastly destroyed the excess memory to avoid leaks.
Output	
Observation	

```
//Loop A
    for(int i = 0; i < j; i++){
        Student* ptr = new Student (nameList[i], ageList[i]);
        studentList[i] = *ptr;
    }
```

```
//Loop B
    for(int i = 0; i < j; i++){
        studentList[i].printDetails();
    }
```

```
Constructor Called.
Constructor Called.
Constructor Called.
Constructor Called.
Copy Assignment Operator Called.
Constructor Called.
Copy Assignment Operator Called.
Constructor Called.
Copy Assignment Operator Called.
Constructor Called.
Copy Assignment Operator Called.
Constructor Called.
Copy Assignment Operator Called.
Carly 15
Freddy 16
Sam 17
Zack 18
Cody 19
Destructor Called.
Destructor Called.
Destructor Called.
Destructor Called.
Destructor Called.
PS C:\Users\TIPQC\Desktop\ACALA CPE21S2\PRELIMS\AT 2>
```

Table 2.3 Final Driver Program

Modifications	

Observations	

Table 2.4 Modification/Corrections Necessary

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C: Solve programming problems using dynamic memory allocation, arrays and pointers
Jenna's Grocery List
Apple PHP 10 x7
Banana PHP 10 x8
Broccoli PHP 60 x12
Lettuce PHP 50 x10
Jenna wants to buy the following fruits and vegetables for her daily consumption. However, she needs to distinguish between fruit and vegetable, as well as calculate the sum of prices that she has to pay in total.

1. Problem 1: Create a class for the fruit and the vegetable classes. Each class must have a constructor, deconstructor, copy constructor and copy assignment operator. They must also have all relevant attributes (such as name, price and quantity) and functions (such as calculate sum) as presented in the problem description above.
   Fruit:

```
class Fruit {
    private:
    std::string name;
    float price;
    int quantity;

public:
    //constructor
    Fruit(std::string n = " ", float p = 0.0, int q = 0)
    name = n;
    price = p;
    quantity = q;

    //destructor
    ~Fruit() {}

    //Copy constructor
    Fruit(const Fruit& other)
    {
        name = other.name;
        price = other.price;
        quantity = other.quantity;
    }
```

```
    //copy assignment operator
    Fruit& operator=(const Fruit& other)
    {
        if (this != &other)
        {
            name = other.name;
            price = other.price;
            quantity = other.quantity;
        }
        return *this;
    }

    //calculate total price
    float calculateTotalPrice() const
    {
        return price * quantity;
    }

    //Attribute display
    void displayAttributes() const {

        std::cout << "Category : Fruit" << std::endl;
        std::cout << "Name: " << name << std::endl;
        std::cout << "Price: " << price << std::endl;
        std::cout << "Quantity: " << quantity << std::endl;
        std::cout << "Total Price: " << calculateTotalPrice() << std::endl
    }
};
```

Vegetable:

```cpp
class Vegetable {
    private:
    std::string name;
    float price;
    int quantity;

public:
    //constructor
    Vegetable(std::string n = " ", float p = 0.0, int q = 0)
    name = n;
    price = p;
    quantity = q;

    //destructor
    ~Vegetable() {}

    //Copy constructor
    Vegetable(const Vegetable& other)
    {
        name = other.name;
        price = other.price;
        quantity = other.quantity;
    }
```

```
    //copy assignment operator
    Vegetable& operator=(const Vegetable& other)
    {
        if (this != &other)
        {
            name = other.name;
            price = other.price;
            quantity = other.quantity;
        }
        return *this;
    }

    //calculate total price
    float calculateTotalPrice() const
    {
        return price * quantity;
    }

    //Attribute display
    void displayAttributes() const {

        std::cout << "Category : Vegetable" << std::endl;
        std::cout << "Name: " << name << std::endl;
        std::cout << "Price: " << price << std::endl;
        std::cout << "Quantity: " << quantity << std::endl;
        std::cout << "Total Price: " << calculateTotalPrice() << std::endl
    }
};
```

2. Problem 2: Create an array GroceryList in the driver code that will contain all items in Jenna's Grocery List. You must then access each saved instance and display all details about the items.

```
const int FRUITS = 2;
const int VEGETABLES = 2;

Fruit** fruits = new Fruit*[FRUITS];
Vegetable** vegetables = new Vegetable*[VEGETABLES];

fruits[0] = new Fruit("Apple", 10, 7);
fruits[1] = new Fruit("Banana", 10, 8);

vegetables[0] = new Vegetable("Broccoli", 60, 12);
vegetables[1] = new Vegetable("Lettuce", 50, 10);
```

3. Problem 3: Create a function TotalSum that will calculate the sum of all objects listed in Jenna's Grocery List.

```
float TotalSum(Fruit* fruits[], int fruitSize, Vegetable* vegetables[], int vegetable
    float totalSum = 0.0;

    for (int i = 0; i < fruitSize; i++) {
        if (fruits[i] != nullptr)
            totalSum += fruits[i]->calculateTotalPrice();
    }

    for (int i = 0; i < vegetableSize; i++) {
        if(vegetables[i] != nullptr)
            totalSum += vegetables[i]->calculateTotalPrice();
    }

    return totalSum;

}
```

4. Problem 4: Delete the Lettuce from Jenna's GroceryList list and de-allocate the memory assigned.

```
//clean up memory
for (int i = 0; i < FRUITS; i++)
    delete fruits[i];

for (int i = 0; i < VEGETABLES; i++)
    delete vegetables[i];

delete[] fruits;
delete[] vegetables;
```

OUTPUT:

```
Acala Grocery Lists:
Category : Fruit
Name: Apple
Price: 10
Quantity: 7
Total Price: 70
Category : Fruit
Name: Banana
Price: 10
Quantity: 8
Total Price: 80
Category : Vegetable
Name: Broccoli
Price: 60
Quantity: 12
Total Price: 720
Category : Vegetable
Name: Lettuce
Price: 50
Quantity: 10
Total Price: 500
Total Sum: 1370
```

The errors I encountered were mostly related to C++ syntax and pointer usage rather than the OOP concepts themselves. My biggest challenge was remembering the correct syntax for constructors, especially the required curly braces, and understanding when to use single pointers (*) versus double pointers (**) for dynamically allocated arrays.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **CoPilot**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Example Generator and Issue explanation last resort**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **It just help me or give a idea how to fix but not entirely fixing my code i still the fixing but just give me a hint of how to fix it**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

COMPUTER ENGINEERING

QUEZON CITY

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.